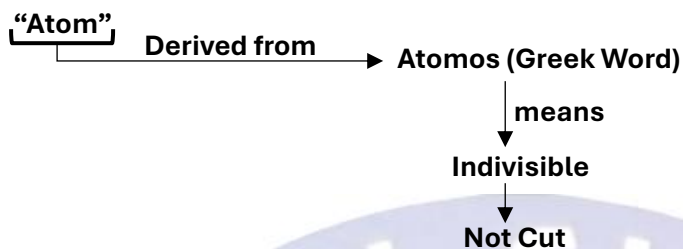


ATOM, MOLECULES & IONS

• Introduction



- According to Indian philosopher "Kannad" [Full name → "Maharshi Kannad" (600 BC)].
"All matters having smallest particle are called "paramanus",

→ Indivisible (Not cut)

The Greek Scientist and philosopher "Democritus and Lucippus" also satisfied by "Maharshi Kannad".
They said that the smallest particle is "atomos",

means → Indivisible (Not cut)

- According to Indian and Greek philosopher, "All substances are made by five basic elements. These are [earth, fire, water, air and sky].

Note:

- All scientists are also interested that why do atoms combine and what happened when they combine.

- According to Antoine Lavoisier atoms combine to form new compounds with new properties.
In 18th Century, Antoine Lavoisier, Joseph Proust, John Dalton and other scientist establishing five important laws of chemical combination.

1. The law of conservation of mass

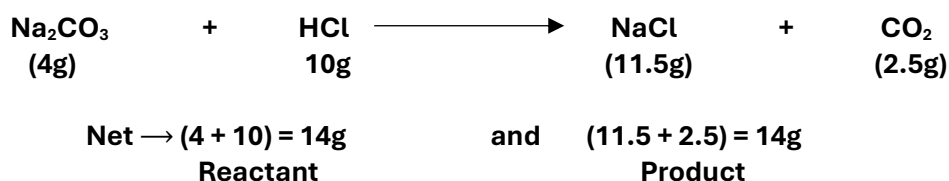
- This law was given by "Antoine Lavoisier in 1774. According to this law, "The total mass of the product of a chemical reaction is equal to the total mass of the reactant that have combined".
- This law is also be defined as, "Mass can neither be created nor be destroyed in a chemical reaction". i.e, mass is conserved in a chemical reaction".

Note:

- This law is also called the "Law of indestructibility of matter".
Experimental example of N.C.E.R.T

- 4g of Sodium carbonate reacts with 10g of HCl then it forms, 11.5g of Sodium Chloride and 2.5g of Carbon dioxide.

i.e,



2. Law of Constant / definite proportions

- This law was given by French Chemist “Joseph Proust” in 1779.
- According to this law, “A chemical compound always contain exactly the same proportion of element by weight / mass”.

Explanation

In H_2O the ratio of H : O $\rightarrow 2^1 : 16^8 \rightarrow 1 : 8$ (by weight)

In $2 \text{H}_2\text{O}$ the ratio of H : O $\rightarrow 4 : 32 \rightarrow 1 : 8$

In $3 \text{H}_2\text{O}$ the ratio of H : O $\rightarrow 6 : 48 \rightarrow 1 : 8$

In CO_2 , the ratio of C and O by weight
 $= 12 : 32$
 $= 3 : 8$

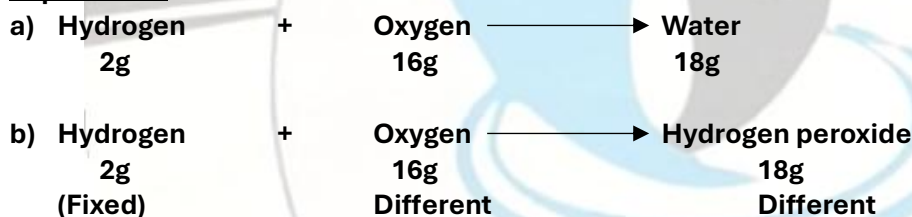
Note:

- The law of constant proportional is only applicable for the pure compound.
- If some impurities are present then ratio may be changed under the condition. Thus, this law can't be apply.

3. The law of multiple proportional

- This law was given by John Dalton in 1803, According to this law, “If two elements can combine to form more than one compound, the masses of one element that combine with a fixed mass of the other element, are in the ratio of small whole numbers”.

Explanation

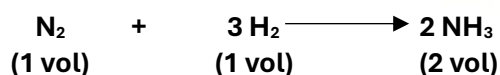
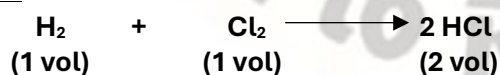


Here, The masses of Oxygen 16g and 32g which combine with a fixed mass of hydrogen (2g) bear a simple ratio. i.e, $16 : 32$ ($1 : 2$)

4. Gay Loussac's law of Gaseous volume

- This law was given by Gay Loussac in 1808.
- “When gases combine or, are product in a chemical reaction they do so in a simple ratio by volume provided all gases are at same temperature and pressure”.

Explanation



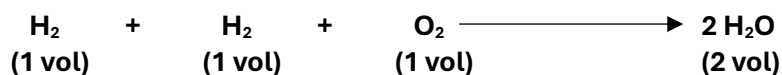
Note:

- This law is actually the law of constant proportion by volume.

5. Avogadro law

- According to Lorenzo Romedo Amedo Carlo Avogadro, “Equal volume of gases at the same temperature and pressure should contain equal number of molecules”.

Explanation



- **Dalton's atomic theory**

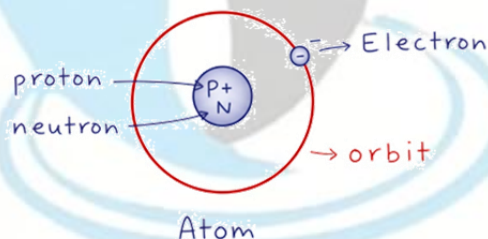
- In 1808, John Dalton gave a famous atomic theory of matter. This is known as, Dalton's atomic theory".
- This theory is basis of law of conservation of mass and law of constant proportional.

- **Assumptions of Dalton's atomic theory**

- Every matter is made up of very small particles known as atoms.
- Atoms are the ultimate particles of matter and can't be further sub – divided into smaller particles.
- Atoms can neither be created nor be destroyed during a physical change or, a chemical reaction.
- All atoms of a particular element are identical in all respect. This means that they have same mass, size, and also same chemical properties.
- Atoms of different element have different masses, sizes and also chemical properties.
- Atoms are the smallest particle of matter can take part in a chemical combination.
- Atoms of the same or, different element combine in small whole number ratio to form molecules of a compound like as, (1 : 1, 1 : 2, 1 : 3, 2 : 3, etc).

- **Demerits / Drawbacks of Dalton's atomic theory**

- According to Dalton Atomic theory atoms are indivisible but atoms are divisible and made up of three fundamental particle: proton, electron, neutron.



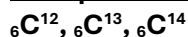
- According to Dalton's atomic theory, same element having same in all respect but it is not true, Atoms of some elements having different mass which is known as Isotope.

For example:-

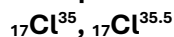
- Isotope of Hydrogen**



- Isotope of Carbon**



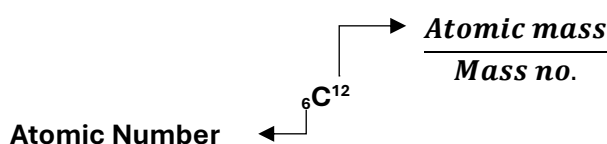
- Isotope of Chlorine**



- ${}_7\text{N}^{14}, {}_7\text{N}^{15}$**

- ${}_{92}\text{U}^{235}, {}_{92}\text{U}^{237}$**

Note:



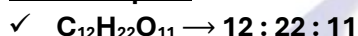
- c) According to Dalton's atomic theory different elements having different atom in all respect but it is not true, different atoms of different elements may same mass which is known as isobar.

For example:-

- i) ${}_6\text{C}^{14}$, ${}_7\text{N}^{14}$
 ii) ${}_{18}\text{Ar}^{40}$, ${}_{19}\text{K}^{40}$, ${}_{20}\text{Ca}^{40}$
 iii) ${}_{27}\text{Co}^{59}$, ${}_{28}\text{Ni}^{59}$

- d) According to Dalton's atomic theory atoms are combine in small whole number such that (1 : 1), (1 : 2), (1 : 3), (2 : 3) but in some cases. It is not true.

For example:-



• **Isotone**

- Atoms of same neutron are called isotones.

For example:-



Here, C and N are isotones.

Trick

<u>Iso, tope</u> Same Atomic no.	Same (Z)	Different	Different
<u>Iso, bar</u> Same mass no.	Different	Same (A)	Different
<u>Iso, tone</u> Same neutron no.	Different	Different	Same (N)



• **Atom**

- Atom is the smallest particle of matter (element / compound). Which can take part in a chemical reaction.
- It may / may not exist in free state.

Exception

- ✓ Inert gas: like as – Helium (He), Neon (Ne), Argon (Ar), Krypton (Kr), Radon (Rn), Xenon (Xe) exist in Free State.
- ✓ Shape of atom is spherical (Hypothetical).
- ✓ Radius of atom of different element are different.
- ✓ It can't be seen by naked eye or, simple microscope.
- ✓ The radius of an atom is measured in nano metre.
- ✓ The diameter of an atom is $1 \times 10^{-10} + 0.5 \times 10^{-10}$ meters (Approx).
- ✓ Atoms can be seen only by Scanning electron machine (S.E.M), electron tunneling machine (E.T.M).

• Atomic symbol

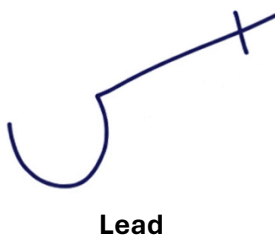
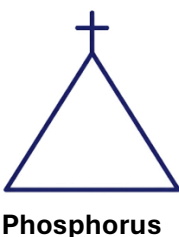
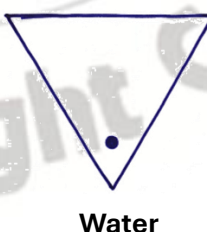
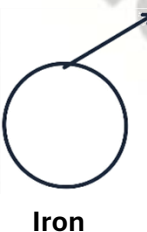
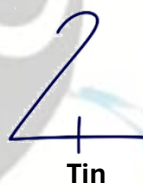
- The shortest names of the element are known as atomic symbol.
- A symbol is an abbreviation / full name of an element.

▪ There are two types of symbols

1. Alchemists Symbol.
2. Modern Symbol.

1. Alchemists Symbol

- Alchemists were the ancient chemist also called Fire philosopher, According to this symbol they used certain sign / symbol based on mythological origin to represent some of the element.



- Dalton's symbols for a few element and compounds are given

Hydrogen	
Oxygen	
Nitrogen	
Carbon	
Sulphur	
Lead	
Copper	
Zinc	
Mercury	
Water	
Carbon dioxide	
Hydrogen disulphide	
Phosphorus	
Silver	
Gold	

Note:

- This symbol suggested alchemists and Dalton's had no logic also not systematic.

2. Modern Symbol

- **Modern Symbol for the elements introduced by J.J Berzelius.**
→ John Jacob Berzelius
- **A symbol is short hand representation of name of an element.**

Note:

- If two or more elements having same 1st letter than we select any other principal letter in the element.

<u>Name of element</u>	<u>Symbol</u>
Boron	B
Beryllium	Be
Bromine	Br
Bismuth	Bi
Barium	Ba

- Some elements there atomic no.(atm) symbol and atomic weight (at wt)

At. No.	Element	Symbol	At.wt Atomic mass unit (a.m.u)
1	Hydrogen	H	1
2	Helium	He	4
3	Lithium	Li	6
4	Beryllium	Be	9
5	Boron	B	11
6	Carbon	C	12
7	Nitrogen	N	14
8	Oxygen	O	16
9	Fluorine	F	19
10	Neon	Ne	20
11	Sodium (Natrium)	Na	23
12	Magnesium	Mg	24
13	Aluminum	Al	27
14	Silicon	Si	28
15	Phosphorus	P	30
16	Sulphur	S	32
17	Chlorine	Cl	35.5
18	Argon	Ar	40
19	Potassium (Kalium)	K	39
20	Calcium	Ca	40
21	Scandium	Sc	45
22	Titanium	Ti	48
23	Vanadium	V	51
24	Chromium	Cr	52
25	Manganese	Mn	55
26	Iron (Ferrum)	Fe	55.8
27	Cobalt	Co	59
28	Nickel	Ni	58.6
29	Copper (Cuprum)	Cu	63.5
30	Zinc	Zn	65.3
47	Silver (Argentum)	Ag	101.9
79	Gold (Aurum)	Au	197
80	Mercury (Hydrargyrum)	Hg	200.5
82	Lead (Plumbum)	Pb	207
51	Antimony (Stibium)	Sb	121.7
74	Tungsten (wolfram)	W	184

Note:

- The symbol of an element indicates the following:-
 1. It represents one particular element.
 2. It represents one atom of the element.
 3. It represents one atomic mass of element.

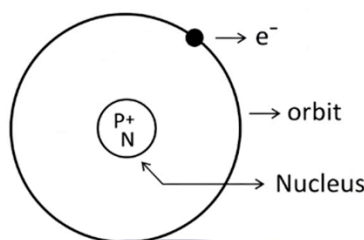
- Atomic mass**

- It is the mass of an atom completely.

i.e, Atomic mass = mass of proton + mass of neutron + mass of electron

$$\Rightarrow \text{Atomic mass} = M_p + M_n + M_e$$

$$= \text{mass of nucleus} + M_e$$



• **Mass number**

- It is defined as, "The sum of mass of proton and mass of neutron".
 i.e, the mass no. is actual mass of neutron.

$$\therefore \text{Mass no.} = \text{no. of proton} + \text{no. of neutron}$$

- Mass no. of an atom is denoted by 'A'.

$$\therefore \boxed{A = Z + N} \quad \Rightarrow Z = A - N$$

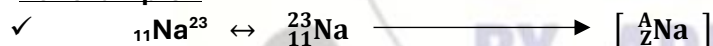
$$\Rightarrow N = A - Z$$

Atoms are electrically neutral because they contain an equal number of positively charged protons and negatively charged electrons.

Note:

1. Atomic no. = no. of proton
 = no. of +ve charge
2. Mass no. $\approx$ Atomic mass
 $\rightarrow$ Approximately equal
 Because, mass of $e^- \approx 0$
3. No. of proton = no. of electron in any atom.

For example:-



$$\therefore N = A - Z \Rightarrow N = 23 - 11 = 12$$

$\therefore$ No. of neutron in sodium (Na) = 12

• **Atomic mass of Relative mass**

- Relative mass of an element is defined as, "It is mass of 1 atom of element and mass of an atom of the hydrogen".

$$\therefore \text{Relative mass of an element} = \frac{\text{mass of 1 atom of the element}}{\text{mass of 1 atom of the hydrogen}}$$

- It can also be simply represented as 'u' (Unified mass).

i.e, atomic mass of Hydrogen = 1 u

atomic mass of Oxygen = 16 u (as compared to Hydrogen)

Note:

- Firstly Hydrogen was considered because Hydrogen was lighter element but problem with hydrogen was that exists in three forms with different masses. So, Hydrogen was replaced Oxygen. But some problem created.

- In 1961, According to I.U.P.A.C most abundant Isotope of Carbon with mass 12 u.

Now, Relative atomic mass of an element = $\frac{\text{mass of atom of the element}}{\frac{1}{12} \times \text{mass of 1 atom of C-12}}$

• **An atomic mass unit**

- The atomic mass unit (a.m.u) is equal to $\left(\frac{1}{12}\right)^{th}$ mass of an atom of C – 12.

$\therefore 1 \text{ a.m.u} = \frac{1}{12} \times \text{mass of C – 12.}$

$\Rightarrow 1 \text{ a.m.u} = 1.66 \times 10^{-24} \text{ g}$
 $= 1.66 \times 10^{-27} \text{ kg}$

- 1 a.m.u (1 u) is also called Dalton (Da).
 ➤ It is also called unified mass (u).
 ➤ It is also called unified atomic mass constant (mu).

Note:

- Atomic masses of elements are their average atomic masses.

• **Gram – atomic mass / Gram atom**

- The atomic mass of an element expressed in gram which is numerically equal to the mass in ‘u’.

For example:-

- ✓ Atomic mass of Oxygen (O) = 16 a.m.u

$\therefore \text{Gram atomic mass} = 16 \text{ g}$
 $= 6.022 \times 10^{23} \text{ atoms of Oxygen}$

▪ **Differences between atoms and molecules**

	Atoms	Molecules
i.	Atom is the smallest particle of an element.	Molecule is smallest particle of substance (element / compound)
ii.	Atom can take part in a chemical reaction.	Molecules can't directly take part in a chemical reaction.
iii.	Atom may or may not exist independently.	Molecule exists independently.
iv.	Atoms having negligible mass and size.	Molecules having masses.
v.	Spherical in shape.	Many shapes – linear, angular and triangular.
vi.	It can't be seen by naked eye or, simple microscope.	It can be seen by microscope.
vii.	Highly reactant exception – Inert gas.	Less reactive.
viii.	No any bond in atom.	Bond between atoms.

- According to John Dalton the smallest particle of a compound is called a compound atom (molecule), but later called molecule according to Italian chemist “Amedeo Avogadro”.

- **Molecule OR (compound atom)**

- Molecule represents a group of two or more atoms (same or different) chemically bonded to each – other and held tightly by – strong attractive forces.
- Molecules are represented in terms of symbols of constituting atoms and it is known as chemical formula.
- The smallest particle of matter (element / compound) which can exist free state is called a molecule.

- **Properties of molecules**

- i. The molecules of any substance are all similar.
- ii. The molecules of different substances are different.
- iii. The properties of a substance are the properties of its molecules.

- **Elementary Molecules**

- Those molecules of an element are made up of similar / identical atoms.

For example:-

- ✓ $H_2, O_2, N_2, O_3, Cl_2, F_2, P_4, S_8, C_{60}$ etc.

$C_{60} \rightarrow$ Giant C – atoms molecules
 → Like football type structure

Name – Buckminster fullerence
OR, Bucky ball
OR, fullerence

- **Types of molecules**

- a) Molecules of elements.
- b) Molecules of compound.

- a) **Molecules of elements**

- Molecules of elements are formed by the combination of two or, more atoms of same element.

For example:-

- ✓ Cl_2, H_2, N_2, P_4 etc

- **Atomicity**

- The number of atoms present in the molecules of an element or compound represent its atomicity.

- **Types of molecules on the basis of atomicity**

- i) **Mono atomic molecules**

- The molecules of atomicity one, called monoatomic molecules.

For example:-

- ✓ All inert gases: He, Ne, Ar, Kr, Xe and Rn are monoatomic exist in free state.

- ii) **Diatomic molecules**

- The molecules of atomicity two, called diatomic molecules.

For example:-

- ✓ $O_2, H_2, N_2, Cl_2, CO, HCl, NaCl$ etc.

iii) Triatomic molecules

- The molecules of atomicity three, called triatomic molecules.

For example:-

- ✓ H_2O , O_3 , CO_2 , SO_2 , H_2S etc.

iv) Tetraatomic molecules

- The molecules of atomicity four, called tetraatomic molecules.

For example:-

- ✓ P_4 , C_2H_2 etc.

v) Polyatomic molecules

- The molecules of atomicity or more than four, called polyatomic molecules.

For example:-

- ✓ S_8 , $\text{C}_6\text{H}_{12}\text{O}_6$, C_{60} , H_2SO_4 etc.

Note:

- All molecules are not called compounds but all compounds are molecules.

For example:-

- ✓ N_2 , O_2 , O_3 , H_2 , P_4 , S_8 , C_{60} etc are called molecules but not called compound.
✓ H_2O , CO_2 , H_2SO_4 , H_2S etc are called compound as well as molecules of compound.

• Molecular Mass Or, Molar mass

- The molecules mass of a molecule is defined as “The sum of at mass of each atom present in a molecule”.

For example:-

- ✓ The molecular mass of ammonia

i) $\text{NH}_3 = \text{At mass of N} + 3 \times \text{mass of H}$
 $= 14 + 3 \times 1 = 17\text{u}$

ii) Molar mass of $\text{SO}_2 = \text{At mass of S} + 2 \times \text{at mass of O}$
 $= 32 + 2 \times 16$
 $= 32 + 32 = 64\text{u}$

• Gram molecular mass

- The molecular mass of the substance expressed in gram which is numerically equal to the molecular mass in u.

For example:-

- ✓ Gram molecular mass of $\text{NH}_3 = 17\text{ g}$
✓ Gram molecular mass of $\text{SO}_2 = 64\text{ g}$

- Gram molecules mass is also called One gram molecule.

Note:

- Gram molecular mass of $\text{SO}_2 = 6.022 \times 10^{23}$ molecules of SO_2 .

- **Formula Unit mass**

- It is just like a molecular mass of substance. But it is only applicable for ionic compound / molecule.

Only for Understand

- We can use either molecular mass or formula mass unit or both in ionic molecule / compound but if ions are not present then we use only molecular mass.

For example:-

- ✓ $\text{NaCl} \rightarrow$ Molecular mass = 58.54
Or Formula mass unit = 58.54

But, $\text{NH}_3 \rightarrow$ Only molecular mass = 17 u

Note:

1. Gram atomic mass of an element contains 6.022×10^{23} atoms.
2. Gram molecular mass of a compound also contains 6.022×10^{23} number of molecules.

- **Explanation**

- The mass of one atom of carbon has been calculated as 1.9924×10^{-23} g.

$$\therefore \text{No. of C - atoms} = \frac{\text{Gram atomic mass of C}}{\text{Mass of one C-atom}} = \frac{12\text{g}}{1.9924 \times 10^{-23} \text{ g}} = 6.022 \times 10^{23} \text{ proved.}$$

- 6.022×10^{23} is known as Avogadro's constant or Avogadro number and it is denoted by 'No' or 'NA'.

- **Mole**

Mole (Latin Word)

↓
Heap, collection Or pile

- A mole denotes Avogadro's number of particles (atoms, ions, molecules, electrons etc).
A mole of atom is a collection of atoms whose total mass is the number of grams equal to the atomic mass.
- Mole is also called Chemist's dozen.

For example:-

- ✓ 1 mole of Nitrogen (N) element = 6.022×10^{23} Nitrogen atoms = 14g.
- ✓ 1 mole of Na element = 6.022×10^{23} Sodium atoms = 23g.

Note:

- ✓ No. of Oxygen atoms in 3 moles = $3 \times \text{No}$
= $3 \times 6.022 \times 10^{23}$
= 1.81×10^{24} atoms

▪ **Moles are expressed in two ways**

Case 1: Expressing number of particles in terms of moles.

$$\text{No. of moles} = \frac{\text{Given no. of particles}}{\text{Avogadro no of particles}}$$

For example:-

Q. Calculate the no. of moles of Fe in a sample containing 10^{22} atoms of Iron.

$$\text{Ans No. of moles} = \frac{1.0 \times 10^{22}}{6.022 \times 10^{23}} = 0.0166 \text{ mol}$$

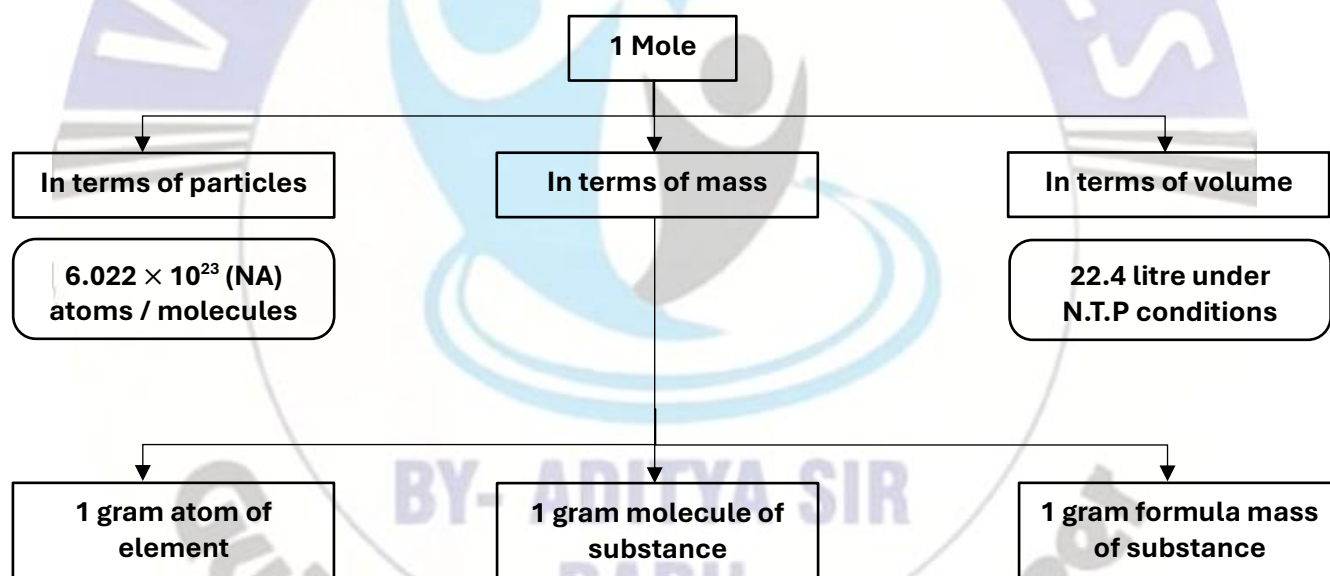
Case 2: Expressing mass of a substance in terms of moles.

$$\text{No. of moles} = \frac{\text{Given mass (in g)}}{\text{Atomic / molecular mass}}$$

For example:-

Q. Calculate the no. of moles 20 g of calcium.

$$\text{Ans No. of moles} = \frac{20}{40} = 0.5 \text{ mol}$$



- According to Avogadro, Equal volumes of all gases under similar conditions of temperature and pressure contain equal number of molecules.
- The normal temperature (0°C or 273°K) and pressure (1 atmosphere or 760mm) also known as N.T.P condition.
- **Molar Volume**
 - One gram mole (or, molar mass) of every gas or vapours occupy a volume of 22.4 litres. This is known as molar volume.
 - Gram molecular mass of gas $\equiv$ Gram Molecular volume.
 - Molar volume = 22.4 litres.

For example:-

- ✓ 2 gm of hydrogen (H_2) occupy 22.4 litre under N.T.P conditions.
- ✓ 32 gm of O_2 occupy 22.4 litre under N.T.P conditions.

▪ **In the case of solutions**

1. **Mass percentage or Weight percentage**

$$= \frac{\text{Mass of solute}}{\text{Mass of solution}} \times 100$$

2. **Mole fraction**

$$= \frac{\text{No. of moles of A}}{\text{No. of moles of solution}} = \frac{n_A}{n_A + n_B}$$

3. **Molarity (M)**

$$= \frac{\text{No. of moles of solute}}{\text{volume of solution in litre}}$$

4. **Molality (m)**

$$= \frac{\text{No. of mass of solute}}{\text{Mass of solvent (in Kg)}}$$

